DSA5204 Project Proposal - Group 19

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Reproducing FlexRAG and Extending to Visual Document RAG on ViDoRe V3 Dataset

1. Publication and project type

The base paper is: Zhuocheng, Z., Feng, Y., & Zhang, M. (2025). FlexRAG: A flexible and comprehensive framework for retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)* (pp. 621–631).

The project type is application. We aim to reproduce and extend existing FlexRAG configurations in two parts. Part A reproduces the FlexRAG pipeline on the same single-hop QA tasks and metrics used in the paper. Part B applies FlexRAG to a new domain - visual document retrieval and multimodal RAG - using the ViDoRe V3 Computer Science dataset (Loison et al., 2026).

2. Introduction to the research problem

Retrieval-Augmented Generation (RAG) systems depend on multiple interacting components - retrievers, indexers, rerankers, generators, and evaluators, making research difficult to reproduce and compare across implementations. FlexRAG (Zhuocheng et al., 2025) addresses this by providing a modular, standardised framework for building and evaluating RAG pipelines, supporting multiple retrieval strategies, pre-configured resources, and evaluation utilities to enable fair comparisons and reproducibility across experimental setups.

FlexRAG evaluates its ModularAssistant pipeline on single-hop QA benchmarks such as Natural Questions (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), and PopQA using a Wikipedia knowledge base, demonstrating that component choice substantially affects downstream quality and efficiency. Our extension is motivated by the fact that these widely used QA benchmarks rely on textual passages derived from Wikipedia, whereas real-world documents such as textbooks, reports, and presentations often contain critical evidence in tables, figures, and diagrams that text-only retrieval may miss. ViDoRe V3 (Loison et al., 2026) is a comprehensive multimodal benchmark spanning 10 document corpora with ~26,000 pages and 3,099 human-verified queries in six languages, with annotations for retrieval relevance, bounding-box localisation, and reference answers.

By applying FlexRAG to the ViDoRe V3 Computer Science subset, comprising 1,360 pages and 215 unique queries from technical textbooks (Loison et al., 2026), we will: (1) stress-test FlexRAG's modularity by adapting it to visually rich documents beyond its published evaluation scope; (2) employ dual evaluation using both ViDoRe's protocol (NDCG@10 for retrieval; LLM-as-Judge for generation) and traditional QA metrics (EM, F1, ROUGE-L metrics not reported in ViDoRe V3), enabling comparison with both visual retrieval baselines and text-based RAG benchmarks.

3. Background reading

To execute the reproduction and extension, the team will read across four areas. First, the FlexRAG paper itself (Zhuocheng et al., 2025), covering its modular architecture, FlexRetriever, FlexReranker, FlexGenerator, evaluation tooling, and resource overhead analysis. Second, retrieval fundamentals - dense retrieval via DPR (Karpukhin et al., 2020), sparse retrieval via BM25S (Lu, 2024), late-interaction models including ColBERT

(Khattab & Zaharia, 2020) and BGE-M3 (Chen et al., 2024), and approximate nearest neighbour infrastructure via Faiss (Douze et al., 2025) and ANN-Benchmarks (Aumüller et al., 2018). Third, context refinement techniques such as RECOMP (Xu et al., 2023) and R4 (Zhang et al., 2024a). Fourth, multimodal and evaluation literature - ViDoRe V3 (Loison et al., 2026), the MTEB framework (Muennighoff et al., 2023), and the RAG survey (Gao et al., 2024).

4. Reproduction and extension plan

Part A - Reproduction

We will reproduce FlexRAG's single-hop QA evaluation on at least one of the datasets - Natural Questions, TriviaQA, and PopQA – using a Wikipedia knowledge base. Our reproduction will follow the ModularAssistant approach, focusing on one method for each component (retriever, reranker, generator). We will deliver a reproduction results table (Exact Match, F1, Success Rate) and briefly compare observed trends against those reported in the paper. The final output of Part A will be a stable, fully reproducible baseline package (configs, seeds, run scripts, and evaluation code) that will be handed off to the extension team.

Part B - Extension

We apply the FlexRAG framework to the ViDoRe V3 Computer Science subset (Loison et al., 2026 [Dataset]), a corpus of two OpenStax computer science textbooks comprising 1,360 pages and 215 unique human-verified queries.

Each ViDoRe page is represented in two parallel views: (i) extracted page text for text-based retrieval and (ii) a rendered page image for visual-aware retrieval; retrieved evidence is referenced using the dataset's page/document IDs.

We first establish text-only baselines (BM25 and dense retrievers) on extracted text and evaluate retrieval using QRELS-based metrics (primary: nDCG@10; optional diagnostics: Recall@k/MRR@k), then evaluate end-to-end QA against the provided reference answers using LLM-as-judge accuracy as the primary metric and ROUGE-L as a lightweight secondary metric. We then evaluate a committed visual-aware baseline leveraging FlexRAG's modular support for multimodal encoders (Zhuocheng et al., 2025 §2.1). Next, we test hybrid text–visual retrieval (e.g., reciprocal rank fusion) and optionally add reranking if compute permits, consistent with ViDoRe V3 findings that visual retrievers and late-interaction reranking can improve performance (Loison et al., 2026). Finally, we test whether FlexRAG's context refiners (Context Repacker, Context Summarizer) improve downstream answer quality. Deliverables: retrieval and end-to-end QA results tables comparing configurations, 4–8 qualitative examples, and a best-configuration recommendation for ViDoRe CS.

5. Evaluation plan

Metric	FlexRAG (Part A - Reproduction)	ViDoRe V3 Computer Science (Part B - Extension)
Retrieval Metrics	Success Rate (Succ)	NDCG@10 (Recall@k + MRR@k as supporting diagnostics)
Generation Metrics	Exact Match (EM), F1 Score	EM, F1 Score, LLM-as-judge & ROUGE-L
Framework/Tool	FlexRAG's internal evaluation	ViDoRe V3's framework (integrated with MTEB leaderboard) for retrieval; LLM-as-judge API + sacreBLEU

6. Division of work

The six-member team is split into two functional subteams plus shared reporting responsibilities. Kalyani Pakhale and Teo Yong Rui own Part A (infrastructure, baseline pipeline, and reproduction experiments). Charles Lim, Adriel Lam, Thiam Wan Xie, and Gabriel Boo own Part B (dataset integration, retrieval evaluation, end-to-end RAG, and extension experiments). All members contribute a methods write-up, and two to three key findings for the final report and presentation. The reading plan and responsibilities are summarised in the table below.

Key Question	Core Concepts	Key Readings	Part	Persons-in-Charge
What is RAG and its core challenges?	RAG architectures, knowledge integration, system complexity	Gao et al. (2024); Zhuocheng et al., 2025 §1	A	Kalyani Pakhale, Teo Yong Rui
How does FlexRAG achieve high performance and low resource usage?	Memory mapping, IVFPQ indexing, async processing, caching	Aumüller et al. (2018); Douze et al. (2025); Zhuocheng et al., 2025 §4	A	Kalyani Pakhale, Teo Yong Rui
How do dense and sparse retrievers work?	Bi-encoders, DPR, BM25, hybrid search	Karpukhin et al. (2020); Lu (2024); Zhuocheng et al., 2025 §2.2.2	A	Kalyani Pakhale, Teo Yong Rui
What are late-interaction models and rerankers?	ColBERT, cross-encoders, T5-rankers, listwise rerankers	Khattab & Zaharia (2020); Chen et al. (2024); Loison et al. (2026) §4.1	B	Charles Lim, Adriel Lam, Thiam Wan Xie, Gabriel Boo
How do you handle multimodal documents?	CLIP, VLMs, visual document retrieval	Radford et al. (2021); Wang et al. (2024); Loison et al. (2026) §1 & 3.3	B	Charles Lim, Adriel Lam, Thiam Wan Xie, Gabriel Boo
How should documents be prepared for a RAG pipeline?	PDF parsing, chunking strategies, context refinement	Zhang et al. (2024a); Xu et al. (2023); Zhuocheng et al., 2025 §2.3.1–2.3.2	B	Charles Lim, Adriel Lam, Thiam Wan Xie, Gabriel Boo
What makes ViDoRe V3 a unique benchmark?	Visual grounding, multi-hop reasoning, multilingual queries, human annotation	Macé et al. (2025); Loison et al. (2026) §2 & 3	B	Charles Lim, Adriel Lam, Thiam Wan Xie, Gabriel Boo
How is RAG performance evaluated?	nDCG, Hit Rate, ROUGE, BERTScore, LLM-as-judge, MTEB	Muennighoff et al. (2023); Kwiatkowski et al. (2019); Joshi et al. (2017)	Both	All

Five-week timeline:

Week 1: FlexRAG setup + sanity checks; confirm reproduction pipeline runs on a small subset.

Week 2: Part A reproduction runs + results table + short reproduction write-up.

Week 3: Part B ViDoRe ingestion + retrieval pipeline setup (text and visual encoders) + retrieval baselines + ViDoRe evaluation.

Week 4: Extension experiments on ViDoRe (visual-aware retrieval using FlexRAG's multimodal encoders; hybrid fusion; text-only vs. visual ablation).

Week 5: End-to-end QA runs + trade-off plots + error analysis + finalize report and presentation.

References (Appendix)

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AI usage declaration

The following AI tools were used in the preparation of this proposal:

(1) **Drafting and editing.** AI assistants were used to restructure and iteratively refine proposal text.

(2) **Citation verification.** AI-assisted web search was used to verify the accuracy of bibliographic details, including author lists, publication venues, years, and DOIs, against primary sources such as arXiv and ACL Anthology.

(3) **Fact-checking.** Specific quantitative claims (e.g., benchmark statistics, evaluation metrics used in cited papers) were verified against the original publications using AI-assisted retrieval.

(4) **All intellectual contributions** - including the project framing, scope decisions, team responsibilities, and technical approach - were made by the group members. AI outputs were reviewed, edited, and approved by the team before inclusion.